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Ahmed Fouad Lagha

AI/ML Engineer & PhD Researcher

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Summary

AI/ML Engineer and PhD researcher building production ML and LLM systems. Shipped two live products and published first-author research on privacy-preserving ML (DP-SGD) and retrieval-augmented generation. Hands-on across the stack (Python, PyTorch, LangChain, vector databases, and FastAPI) with Docker, Kubernetes, and CI/CD in production workflows.

Technical Skills

Programming: Python, SQL, JavaScript/TypeScript, Bash, C/C++
ML & DL: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV
LLM & GenAI: RAG, LangChain, LlamaIndex, Vector DBs
MLOps: Docker, Kubernetes, MLflow, Weights & Biases, Azure, CI/CD
Backend & Serving: FastAPI, REST APIs, WebSockets, vLLM, Ollama
Databases: PostgreSQL, MySQL, MongoDB, SQLite

Professional Experience

Researcher & Teaching Assistant | Eötvös Loránd University — Budapest, Hungary **Sep. 2025 – Present**

- Built and optimized DP-SGD federated-learning pipelines for privacy-preserving synthetic data generation ($\varepsilon = 1$)
- Developed reproducible data preparation, feature engineering, and evaluation workflows for privacy-utility benchmarking
- Teaching Assistant for the Introduction to Data Security Practicum, covering adversarial ML, privacy auditing (membership inference), and DP-SGD/federated-learning defenses

AI Engineer Intern | Monad — Algiers, Algeria **Oct. 2024 – Jul. 2025**

- Contributed to Monad's AI products: a customer-support chatbot, a legal assistant (9anün), and semantic search (3xS)
- Built web scraping and OCR pipelines converting web pages and scanned documents into structured text for training and indexing
- Implemented semantic search and retrieval over document collections using embedding models and vector similarity
- Trained and evaluated NLP models for Narwhal, Monad's social listening product (sentiment, topic, and trend detection)

Software Engineering Fellow | ALX Africa — Remote **Jun. 2023 – Feb. 2025**

- Designed and implemented backend services and RESTful APIs with database integration in Linux environments
- Built modular, maintainable software solutions through iterative development and code reviews

Projects

Vellum — AI-Powered Life Story Platform **2025 – Present**

Next.js · TypeScript · FastAPI · Groq Llama 3.3 · WebSockets · Stripe

- Built and shipped a full-stack AI interview platform (Next.js/TypeScript frontend, FastAPI backend) that turns voice or text interviews into premium PDF life-story books, with NextAuth v5 auth, Stripe one-time payments, and real-time WebSocket chat
- Delivered production-grade engineering: EN/FR internationalization, a 120-test suite (Jest, Playwright, pytest), GitHub Actions CI/CD, Sentry monitoring, and deployment on Vercel and Render

EVA — École Virtuelle Algérienne **2023 – Present**

Next.js · React · Tailwind CSS · i18n · Sentry · Vercel

- Built and deployed a production e-learning platform serving Algerian students across primary, middle (BEM), and high school (BAC), aligned with the official national curriculum
- Implemented multilingual EN/FR/AR interfaces with full RTL layout, course search, learning paths, instructor profiles, and voucher-code offline payments; integrated Sentry for production error monitoring

Publications

- **CASPER: Contextual Anonymized Synthetic Prototypes for Enhanced Retrieval in Medical Information Systems** 2026
  Ahmed Fouad Lagha, Loubna Seddiki, Oumaima Araar, Imre Lendák — *IDEAL 2026, Main Track*
- **Context-Aware Access Control for IoT Devices Using Two-Role LLM Reasoning: A Security Evaluation** 2026
  Ahmed Fouad Lagha, Loubna Seddiki, Oumaima Araar, Imre Lendák — *ITAT 2026, Workshop on Applied Security (WAS)*
- **A Survey of IoMT Datasets: Availability, Applications, and Research Insights** 2026
  Loubna Seddiki, Ahmed Fouad Lagha, Imre Lendák — *IEEE ICCC 2026*

Education

PhD in Informatics | Eötvös Loránd University, Hungary — Sep. 2025 – Expected 2029
M.Sc. in Artificial Intelligence | University of Khenchela, Algeria (GPA: 17.6/20 · Ranked 1st of 95) — Sep. 2023 – Jun. 2025
B.Sc. in Computer Science | University of Khenchela, Algeria (GPA: 16.2/20 · Ranked 1st of 120) — Sep. 2020 – Jun. 2023

Certifications & Awards

Stipendium Hungaricum Scholarship — Fully funded PhD studies at ELTE — Sep. 2025
2nd Place Starthon'25 Hackathon — University of El Oued, Algeria — Apr. 2025
Software Engineering (Back-end Specialization) — ALX Africa — Feb. 2025
AI Career Essentials — ALX AiCE — Jul. 2024

Languages

Arabic (Native), Tamazight (Native), French (Fluent), English (Fluent)